



MELBOURNE
MICROFINANCE
INITIATIVE

Global Microfinance Case Competition 2023

UFH:

*Towards an
Inclusive and
Sustainable
Financial Network*

Team:

The Big Four

Bhavika Rajora

Falak Kalra

Ishan Anand

Parth Khanna

Analysing the landscape

Problems

Solutions

Implementation and Impact

Appendix

SOCIOECONOMIC ANALYSIS



Assessing Indonesia's Macroeconomic Challenges To bridge gaps through proposed strategy

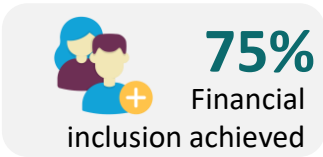
Poverty puts the heaviest pressure in...



Papua, West Papua and West Nusa Tenggara have poverty rates above 20%

The literacy rate is lowest in Papua with approximately 20% of the population over 15 being illiterate

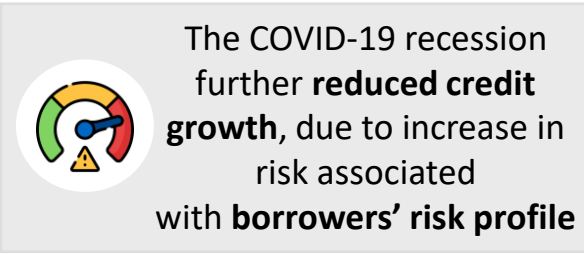
As of 2020



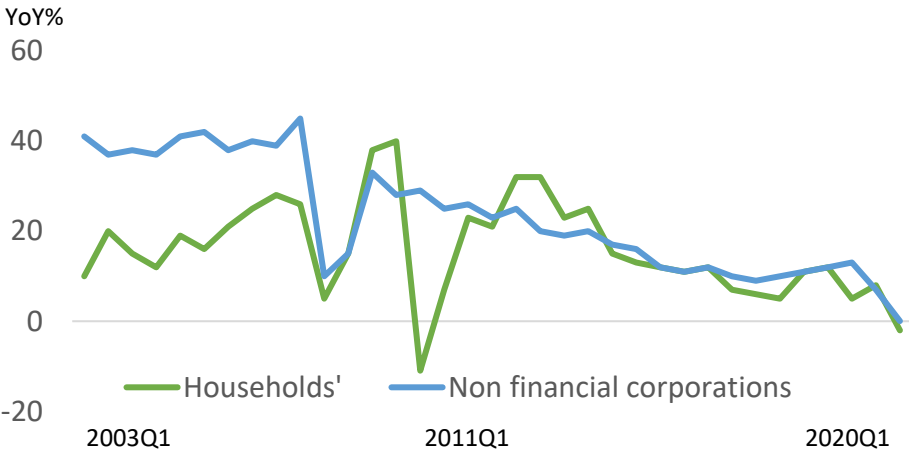
yet



Macroeconomic Indicators



The Way Ahead



Weakened Credit Growth to the Real Sectors

“Economic insecurity coupled with after-effects of Covid-19 has degenerated financial inclusion in Indonesia and the only way out is a decentralised Sustainable Credit Inclusion Grid (CREDILINK)”

Sources: OCED library, ABF, OJK Surveys

EXPLORING THE MICROFINANCE LANDSCAPE IN INDONESIA

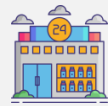
Analyzing Indonesia's financial profile and addressing key challenges through proposed strategy



“ Indonesia is renowned for its large-scale **microfinance sector**, with a range of **commercial banks** and over **60,000 MFIs** reaching more than **50 million** people

174 trillion IDR

Funds distributed
by Rural Banks



50 mn + MSMEs

Contribute more than
30% of GDP growth

Understanding the Financial and Technological Environment



Mandate by Bank Indonesia

Banks to have at least **20%** of their loan portfolio in **micro loans**



Delay in Loan Repayments

Increase in NPLs from 3.35% to **3.95%**

The Mobile – First Nation (Penetration Numbers)

75%

Online purchases are mobile based



270 mn

Projected internet users by 2028

Integral Challenges

Adverse Repayment Abilities of Micro Borrowers

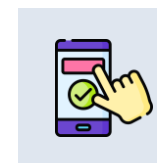
Increased Cost of Borrowing

Arduous Loan Application

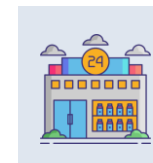
Unspecialised Product Offerings



What can be done?



Utilise
Technological
Advancements



Prioritise **MS**
ME
Requirements



Tap Potential
of Open **API**
Networks



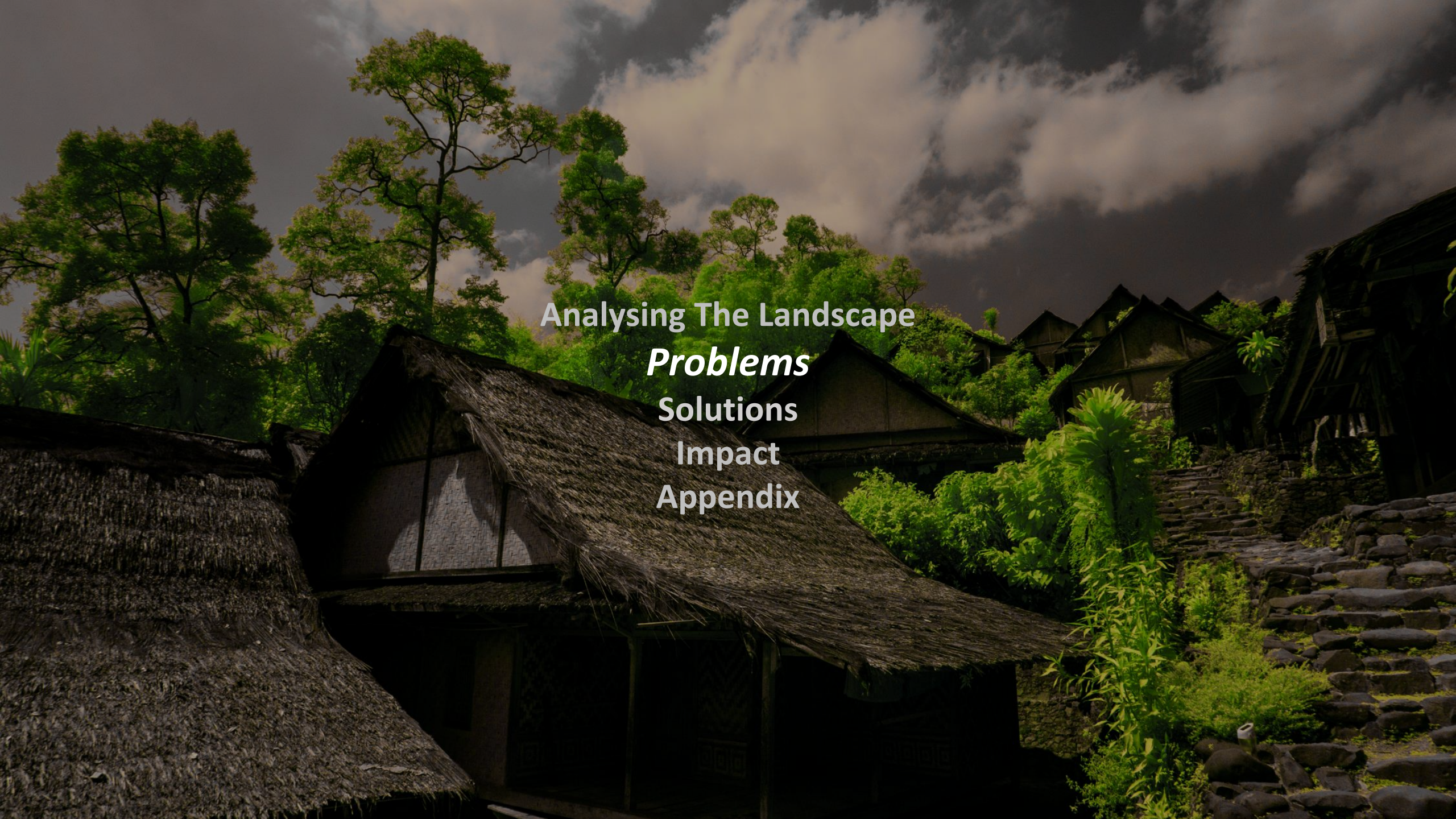
Launch
Targeted **Product**
Offerings



Introduce
Community
based Lending

Strategy Phoenix

Sources: GBG Indonesia, McKinsey reports



Analysing The Landscape

Problems

Solutions

Impact

Appendix

LENDING IN JAVA AND SUMATRA



Arduous Loan Application and Operational Inefficiency backend procedures Chokepoints



Adika, a 28 year old farmer from Java with an annual income of IDR 25.5 mn



Onset of Cropping Season



Need for Fertilizers

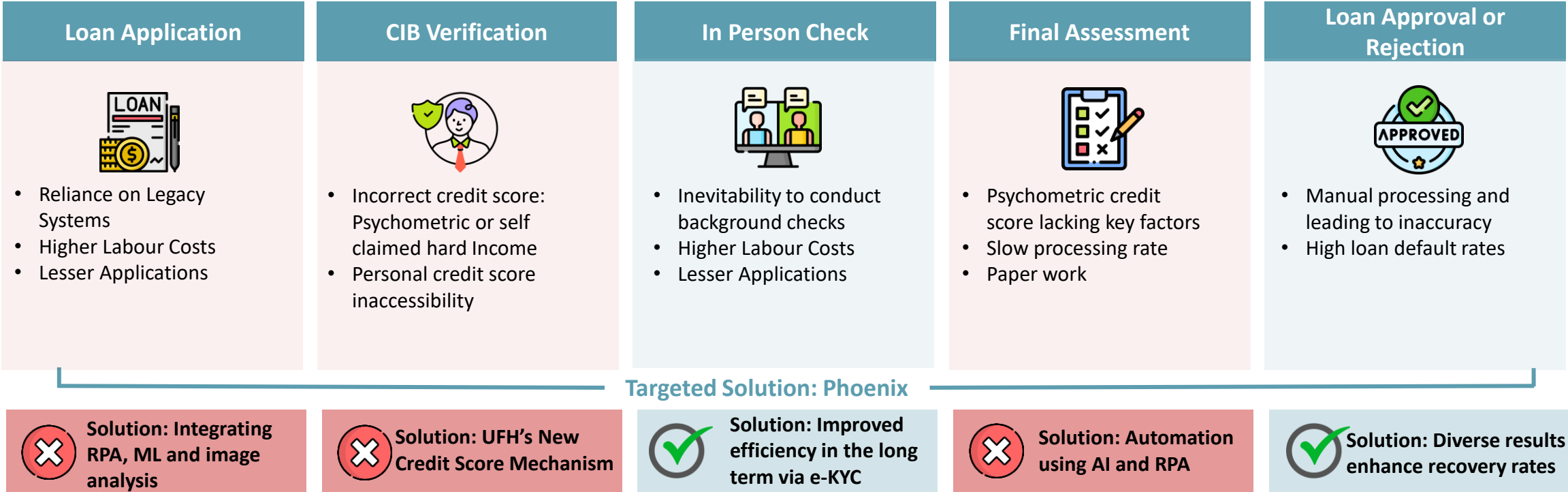


Loan Application In Process for 10 days



Application rejected, Adika borrows from legacy lenders

UFH's Credit Lending Mechanism



Sources: Team Analysis

A scenic view of terraced rice fields in a mountainous region. In the foreground, two people are working with a wooden tray filled with harvested rice stalks. One person is wearing a red shirt, and the other is wearing a blue shirt and a red headscarf. The background shows rolling hills with terraced fields, some of which are green and others are brown, indicating different stages of the rice cycle. The mountains in the distance are hazy and blue.

Analysing The Landscape
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INTRODUCING PHOENIX

Implementation and Impact Assessment through breakdown of Strategy Phoenix



Immediate solutions

Potential solutions or micro-battles	Parameters of Assessment					Overall Attractiveness
	Investment Required (CapEx: Small/Medium/Large)	Degree of Impact	Complexity of Implementation	Duration of Implementation (Rankings 1-5: longest - fastest)	Impact Segment from Identified Choke Points	
Automation and Elimination of Legacy System	Medium	~30% Lesser time taken for the loan acceptance process	Medium	3 rd	Value Chain Inefficiency	
Lending Circles	Small	~800k+ People impacted by imparting Financial Inclusion	Low	4 th	Improved Javan-Sumatran branch collaboration and facilitating rural banking	
Revamping Credit Scores	Small	~98.75% Accuracy in Repayment Prediction	Low	5 th	Falling rates of loan recovery	
Product Diversification	Medium	Increase in Cash Flow by 214 mn IDR	Medium	2 nd	Expansion into Papua and other neighbouring regions	
Credilink: Changing the landscape of Indonesian economy	Large	8% Rise in Applied Loans	High	1 st (longest time)	Driving Fintech and changing landscape	

Sources: Team Analysis



FACILITATING FINANCIAL INCLUSION AND EXPANSION OF UFH



Phased implementation of Lending Circles leading to financial inclusion of 800k+ people in the regions



>> Bridging the gap by **Banking the Unbanked** in Java and Sumatra through the launch of Lending Circles <<



Financial Inclusion

IDENTIFY



Community Lending

BUILD



Impact Generated

IMPLEMENT

[Refer to Appendix](#)

800k +
People directly impacted through Lending Circles

2.36%
Projected YoY Rise in Financial Inclusion

● **Targeting the Underutilized Agriculture Sector**

- **80%** farmers in Java don't own the land
- Employs **50.69%** of the rural population in Sumatra, especially in **palm oil** production

● **The Rising Need for Agricultural Loans**



Decline in Agricultural loans as % of total loans while increase in IDR of the loans, signifying the gap in agricultural credit

● **Applying for the Loan**

- Application Form and Survey
- Financial Education Courses

● **Joining the Group**

- Deciding Loan Amount and Order
- Signing Loan Documents

● **Saving and Building Credit**

- Loan offered at **0% interest rate**
- Payments reported to UFH
- **Credit Score** updated

● **Adopting Dual Model**

- Digital communication in tier 1 cities and towns
- Leveraging existing **agent network** in Papua and NTT with poverty and illiteracy >20%

● **Launch and Expansion**

- Pilot project in Java and Sumatra to facilitate coordination
- Expansion to **Papua** and **East Nusa Tenggara** in Phase 2

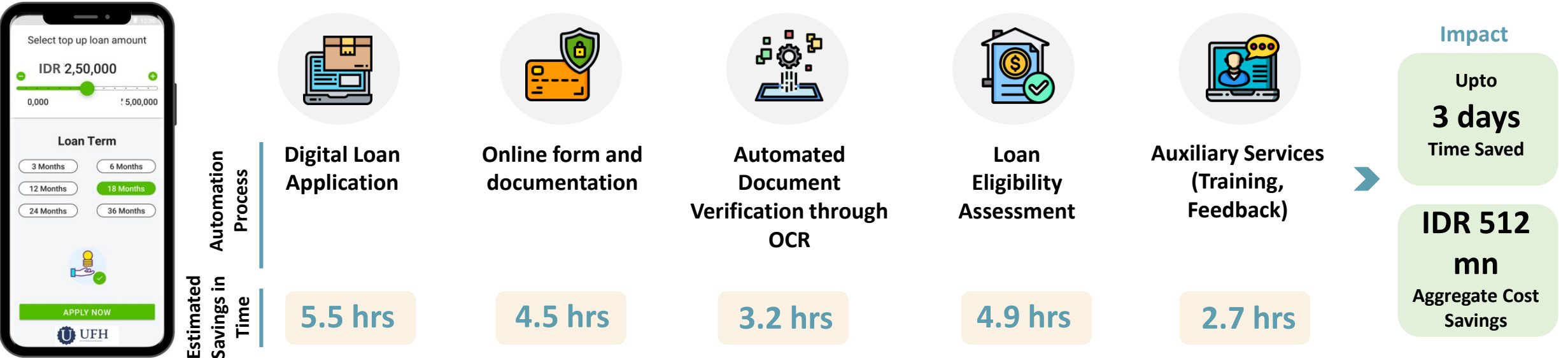
Sources: Mission Asset Fund. IMF, Team Analysis

UFH's CREDIT SCORING

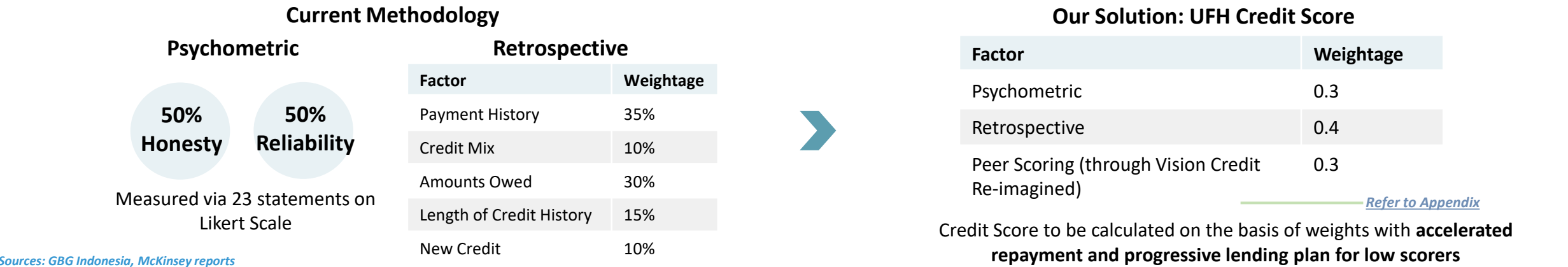
Unveiling The Potential Of Automated Processing And An All-encompassing Credit Score



AI-volution: Optimising the Loan Procedure



Loan Eligibility Assessment: Deep Dive

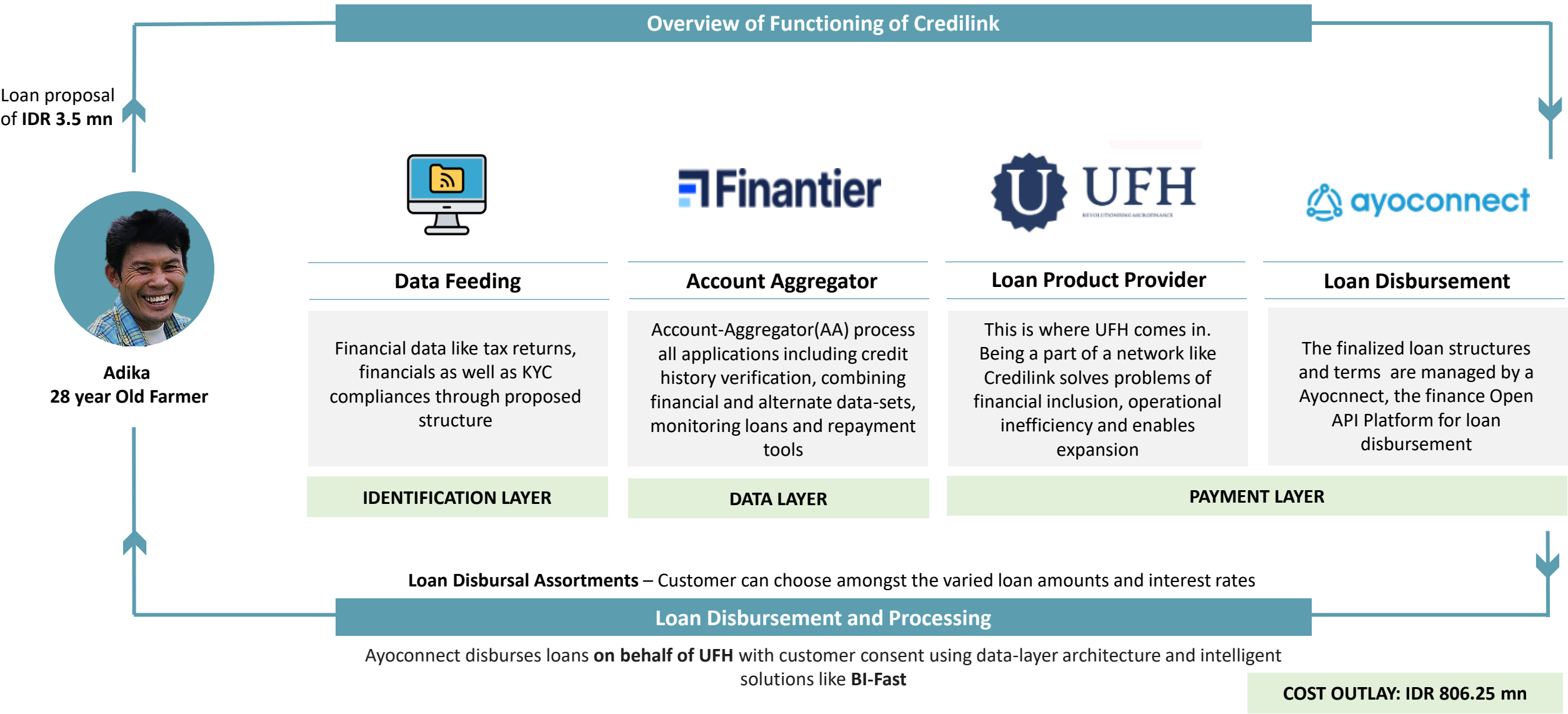


Sources: GBG Indonesia, McKinsey reports

INTRODUCING CREDILINK : THE 3- PRONGED FINANCE (1/4)



Transforming the Micro Loan procedure through



Sources: Finatier's website

[Refer to Appendix](#)

CREDILINK : THE IDENTIFICATION LAYER (2/4)



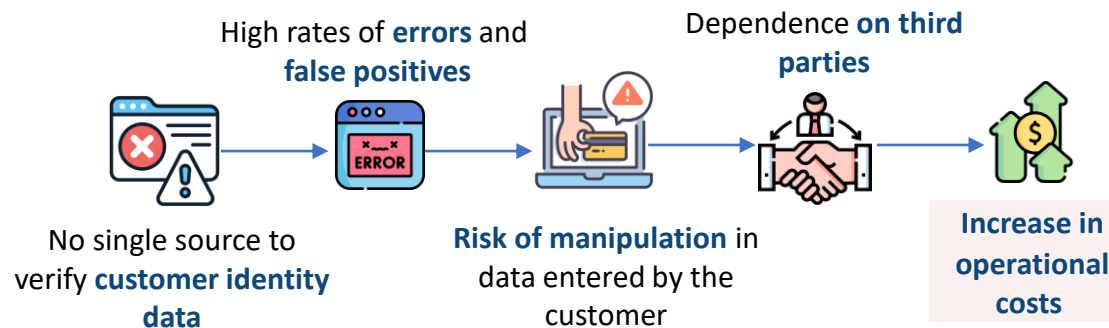
E-KYC: Addressing the Inefficiencies of the Current Credit Lending System by Addressing the Current Players

Structure of Current Credit Lending System

SOURCE	COST	PROCESS TIME
E-money players	IDR 16,000 – 115,000	1-2 Days(customer) 3-10 Days(Merchant)
P2P Lenders	IDR 26,000 – 76,000	1 day(customer) 1-3 Days(Merchants)
Banks	IDR 24,000 – 64,000	2 Months (G2P beneficiary)

Prolonged Loan Approval Duration Leading To Inefficiencies

Assessing the Industry's Value Chain



Sources: MSC Policy

Mapping the Way Forward: Utilising e-KYC

98% of the adult population with **Kartu Tanda Penduduk (e-KTP)** ID card



Feasibility

of digital identity and e-KYC solution for financial inclusion highlighted by GoI



Booming digital economy and fintech sector

USD 3.9-4.2 billion

Savings in Indonesian FinTech using e-KYC

Real Time Biometric Authentication



1 Online Customer onboarding



2 Web transferring of details by agent



5

Successful
Unsuccessful

Authentication

4 Authorised Party Intermediary

6



7 AML + Enhanced Due Dilligence

8 Account Generated

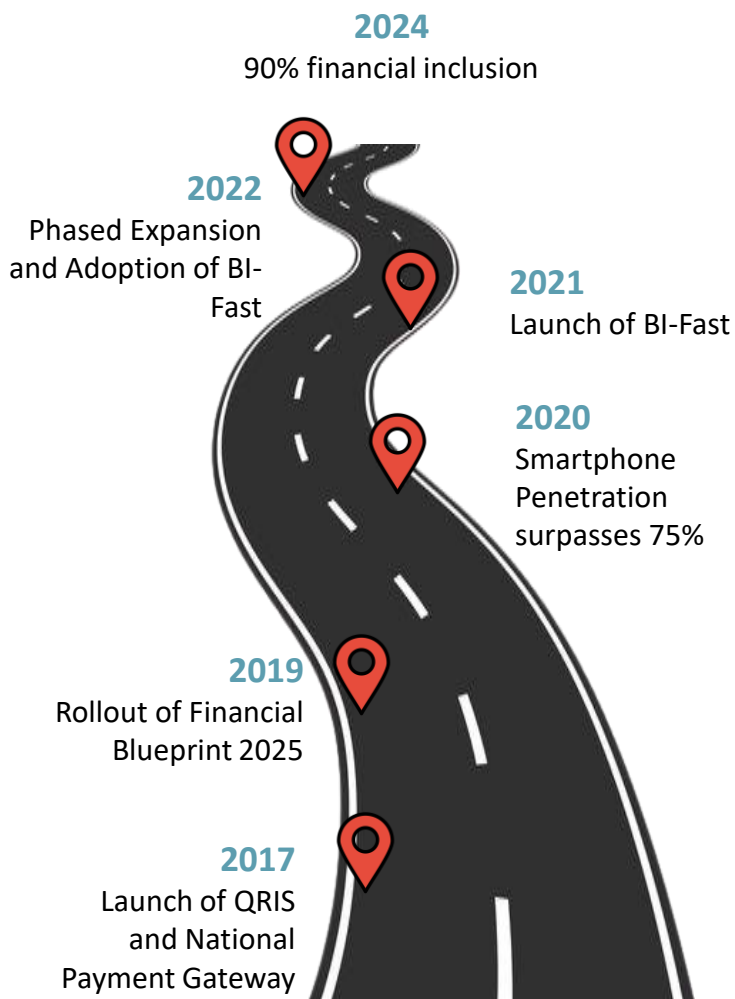
Financial Services Provider (in-house application)

CREDILINK : THE PAYMENT LAYER (3/4)



Moving Away From Legacy Systems And Leveraging The BI – Fast Service For Loan Disbursement

How it all started...

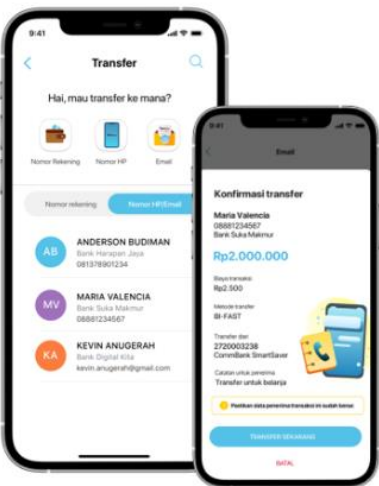


Sources: GBG Indonesia, McKinsey reports

...to AyoConnect Disbursement Solution powered by BI-Fast

Why partner with AyoConnect ?

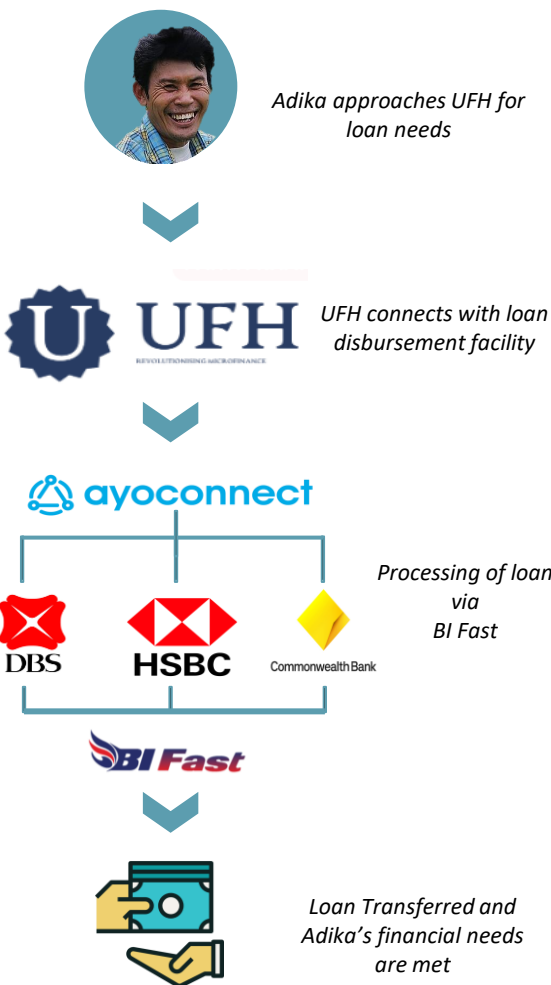
- Licensed by **Bank Indonesia**
- Flexibility to send funds to **70+ banks** in Indonesia that work together in **real time**
- State Of The Art Dashboard** to keep track of payment history and credit scores
- Lower transfer cost than market rate, priced at **IDR 2400 (\$0.16)**



Choice of Payment Service for Loan Disbursement

Payment Service	Real- Time Transfer	Enhanced Security	Low Charges	High Transfer Limit
Online Transfer	✓		✓	
RTGS		✓		✓
BI- Fast	✓	✓	✓	✓
SKNBI		✓	✓	
RTOL	✓	✓		✓

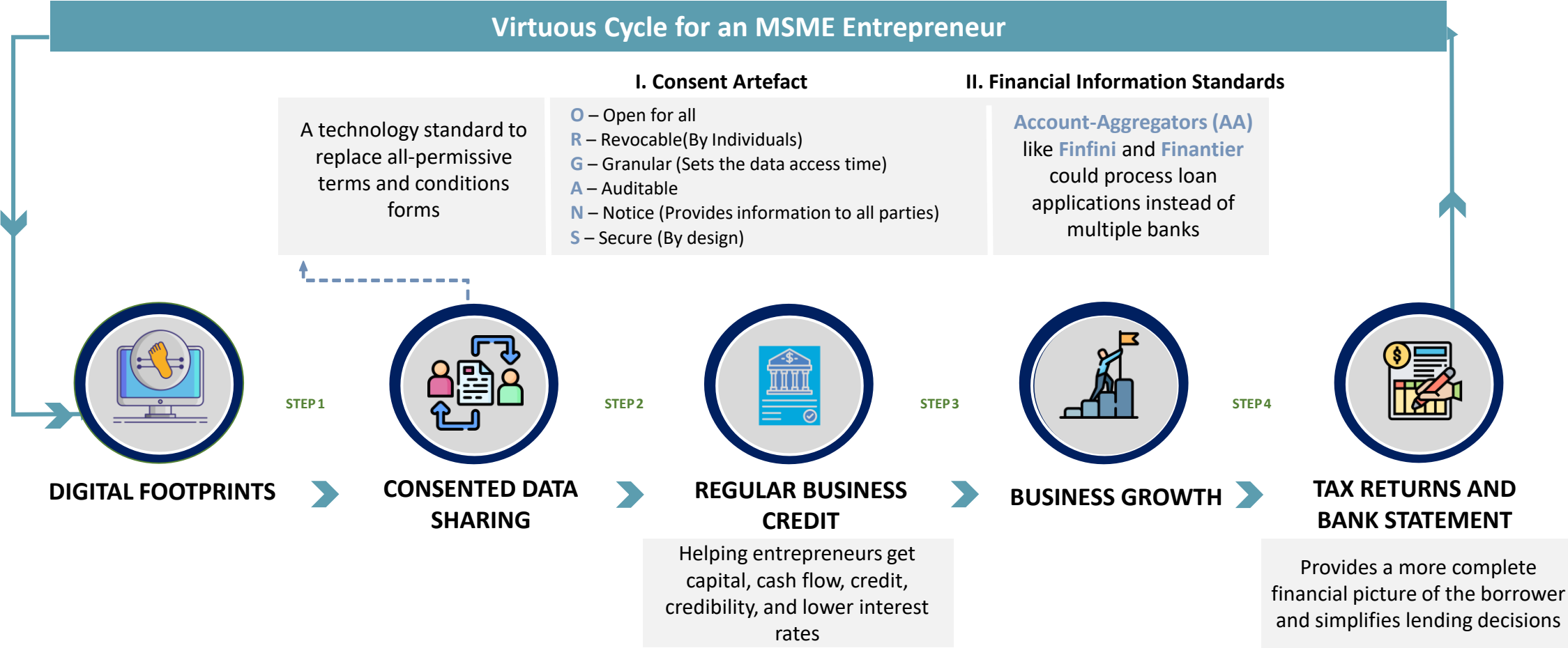
Loan Transfer Journey



CREDILINK: THE DATA LAYER(4/4)



Leveraging Bank Indonesia's BI- FAST Service To Facilitate Real Time Transfer Of Micro Loans

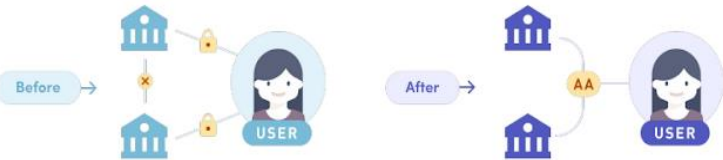


USP of AA Model



Auli is an empowered MSME entrepreneur

Auli can now securely share their financial data with financial institutions in the **AA model**



A DEEPER LOOK INTO THE PRODUCT OFFERINGS



Agriculture Financing enables tapping Unexplored Market Potential and facilitates Expansion

Current Offerings

1051

Individual Loans
Problems solved through Credit Reimagined with improved Financial Inclusion

671

SMEs
Aided through Agricultural Value Chain Loans and Credilink

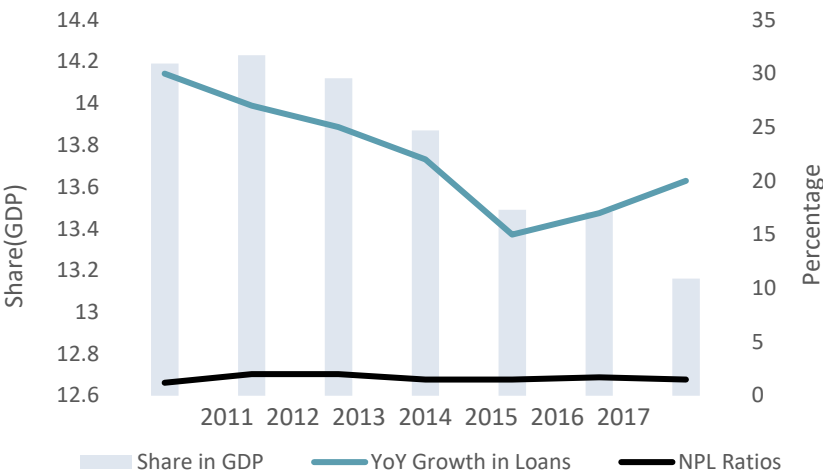
54

Established Loyal Clients
Tapped through Credilink

New Offering: Reasons to expand into Agriculture

Exploring the Demand-Supply Gap

- Relative to GDP Contribution, Agriculture sector is underfunded
- MiBank and Women’s Micro Bank are **the only financial institutions** to have introduced credit products aimed at **rural agricultural communities**



Graph showing the Demand-Supply Gap through underfunding via MFIs in Agriculture

Political Factors

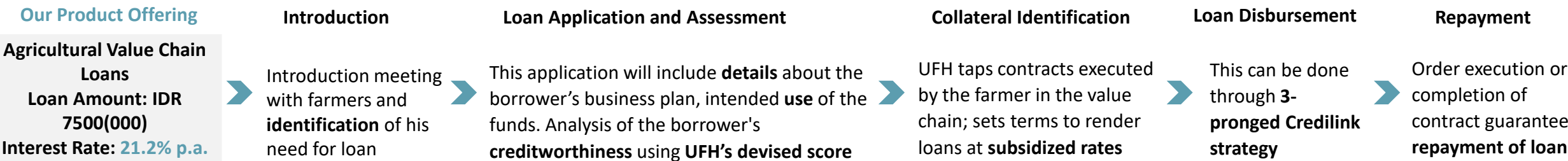
1

Failed Policy Initiatives: Kredit Usaya Rakyat limited to just **26%** or IDR 36.4 trillion of all agricultural commodities



2

Russia- Ukraine War: Effect on prices of fertilisers, a necessary raw material for the agriculture sector



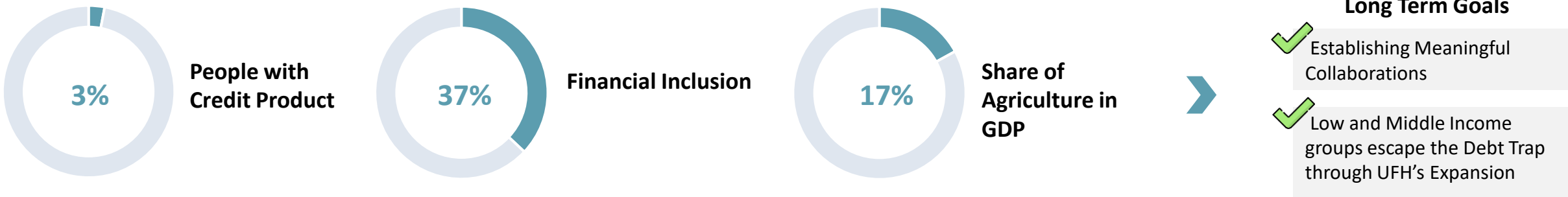
Sources: OJK, OECD Database

SCALING AND EXPANSION



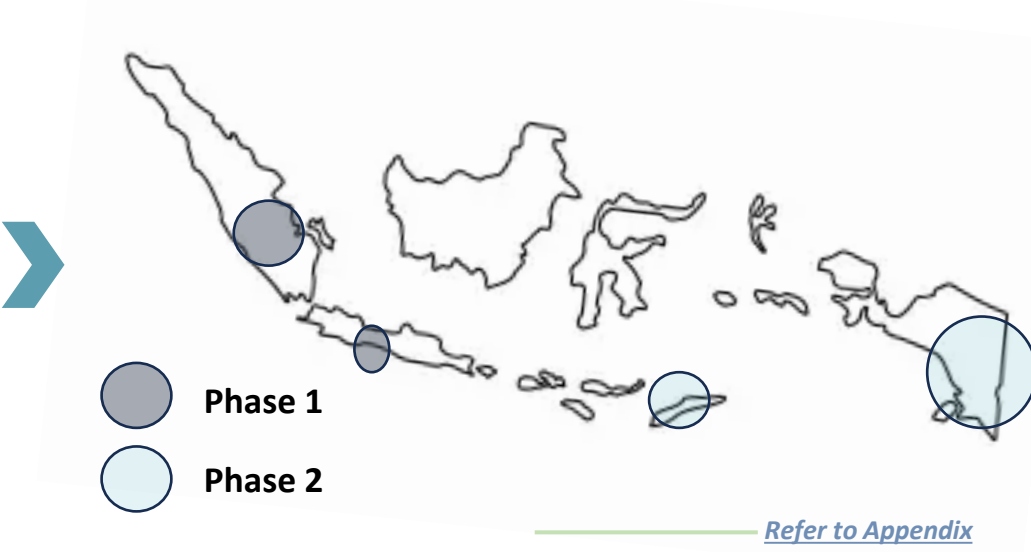
Papua's Untapped Potential: Agricultural Diversification for Sustainable Household and Business Growth

Analysing the lending landscape in Papua and East Nusa Tenggara



Phased Expansion Plan

Phase 1: Java and Sumatra Perfecting the Model: Testing the feasibility, scalability and impact on operational efficiency through short term solutions of Strategy Phoenix Time Frame: 3 yrs Penetration Clientele: 5900+	Phase 2: Papua and East Nusa Tenggara Huge Untapped Market: Over 900 farmer groups, less than 33% utilization capacity and no Agricultural Extension Training centres Time Frame: 3 yrs Penetration Clientele: 6100+	Phase 3: Nation Wide Expansion Making use of lessons learnt from Phase 2 and Credilink to drive a nationwide macroeconomic Time Frame: 2 yrs Penetration Clientele: Pan Indonesia
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Sources: OECD Database, Team Analysis

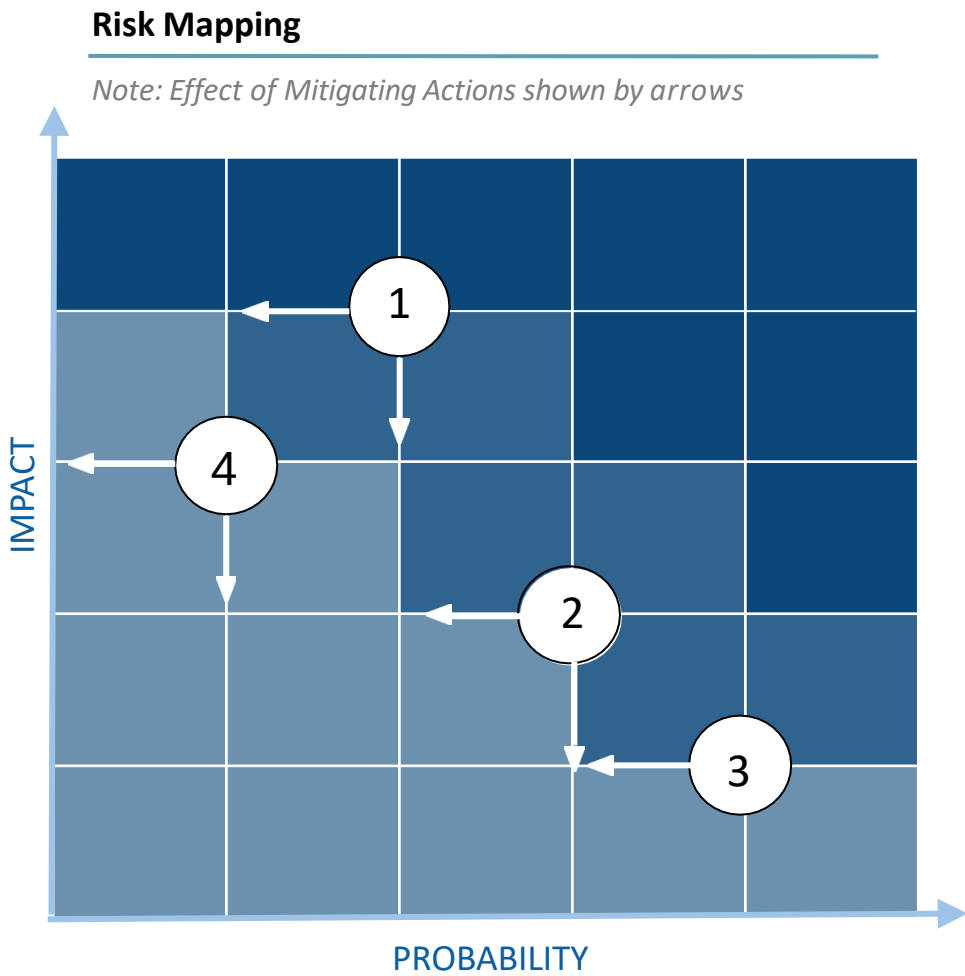
A photograph of three young children standing in front of a building with a corrugated metal roof and a grey brick wall. The child in the center is a boy wearing a blue t-shirt, smiling broadly with his arms around the two children on either side. The child on the left is a girl wearing a grey t-shirt, also smiling. The child on the right is a boy wearing a colorful t-shirt with a graphic of a person and the word 'OSCAR' visible, looking slightly to the side. The background shows some greenery and a clear sky.

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RISK ANALYSIS AND MITIGATION



Suitable Strategies to help UFH preemptively tackle risks due to external and operational factors



Risk	Mitigation	KPIs
1 Update Operational Systems to leverage Technological Advancements	- Invest in cloud computing, big data analytics - Conduct regular training for employees on efficient usage of new technologies	Reduction in Operational Costs
2 Modify or Add to Existing Product offerings to enhance Financial Inclusion	- Conduct market research to understand the needs of low-income Indonesians - Partner with other financial institutions to offer new products	Increase in the number of Low-Income Indonesians who are able to save and invest
3 Restart Collaboration between Sumatra and Java	- Create a shared IT platform - Establish joint marketing initiatives - Create a dedicated team to facilitate collaboration	Increased number of Regional Transactions
4 Introduce Agriculture Financing	Offer loans and savings products that are tailored to the needs of farmers	Increase in Farmers' Income

IMPLEMENTATION TIMELINE

Decoding the step by step formulation and implementation of Strategy Phoenix



Priority wise Collaborations to generate maximum impact...



FINANTIER: *Company solving business problems via Tech*

One step solution for Automation, Credit Scoring and Account Aggregation

1



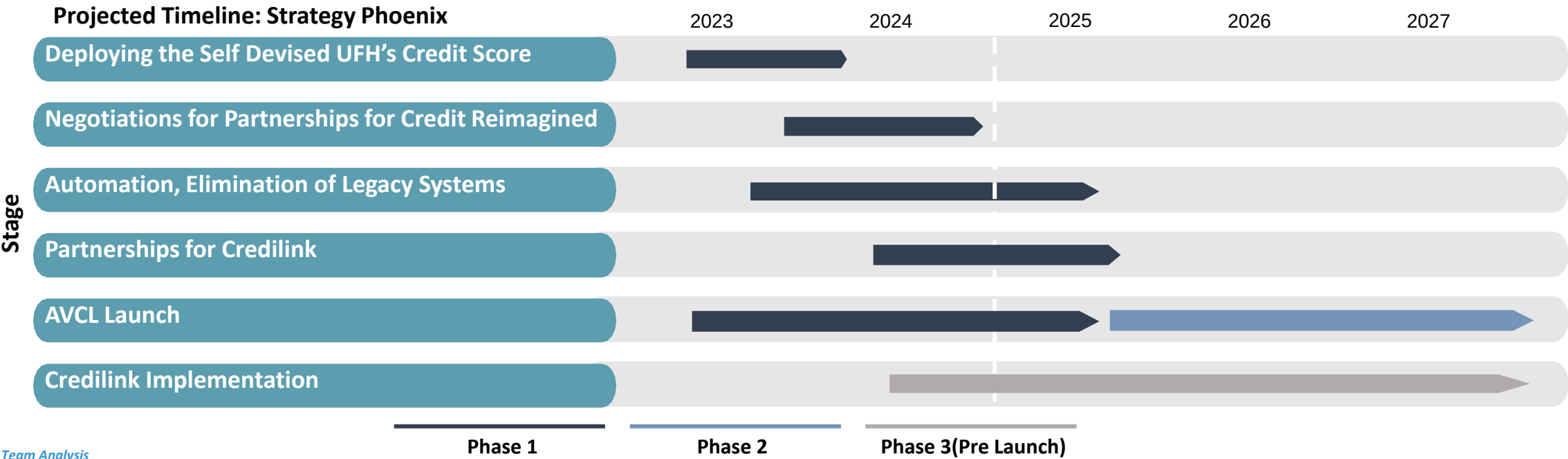
AYOCONNECT: *Company aiding Loan Disbursement*

Integral ending node of the Payment Layer in Credilink Mechanism

2



...Impact creation through Fulfillment of SDGs

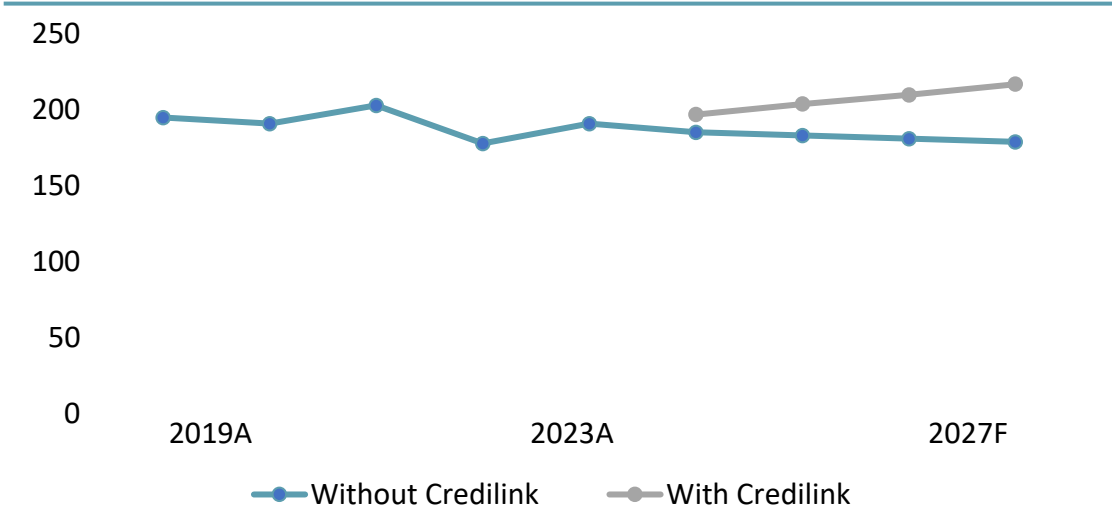


FINANCIAL ANALYSIS OF THE PROPOSED STRATEGY

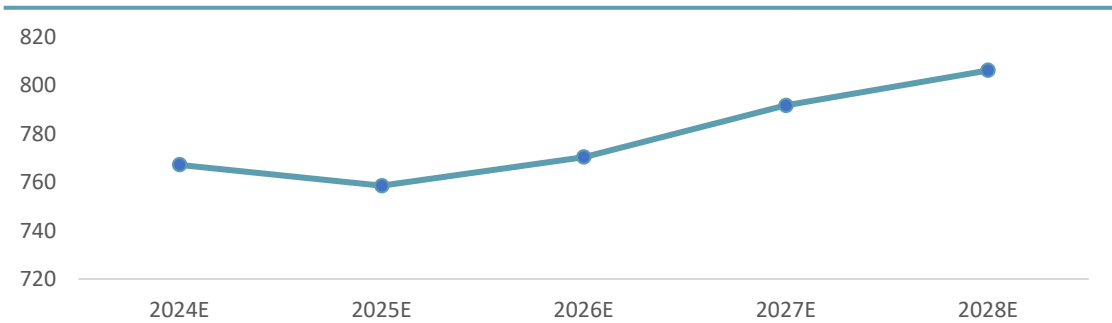


Leveraging the need of the hour to get a high positive NPV through Phoenix

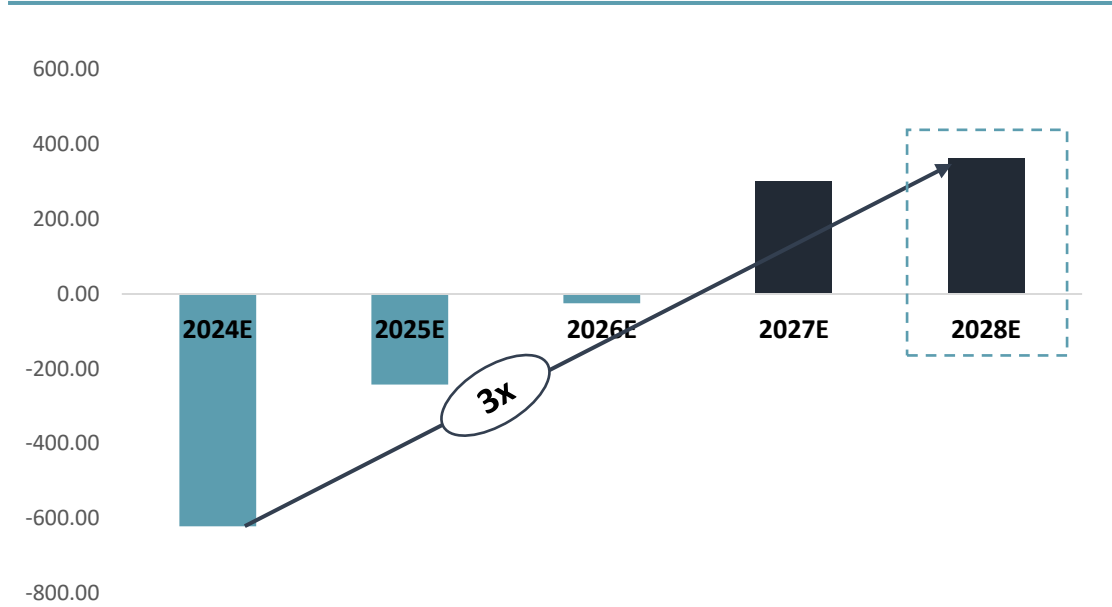
Rise in Loan Acceptance Rate: Increase in Operational Efficiency and Financial Inclusion
Number of Loans Accepted



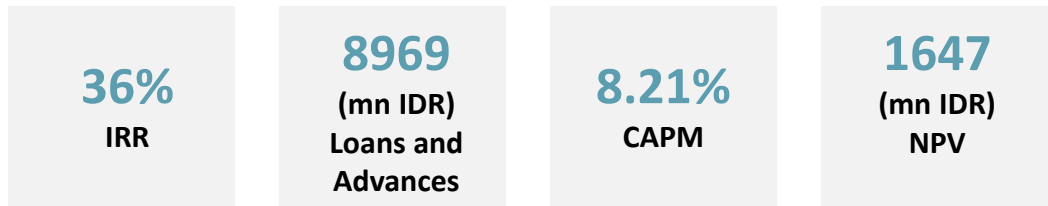
Cost required to develop Credilink
Cost(in mn IDR)



Positive Cash Flows: Viable and Implementable Strategy
Cash Flow Projections(in mn IDR)



Why Strategy Phoenix?



Sources: Team Analysis [Refer to Appendix](#)

A person wearing a hat and dark clothing is bent over, working in a wetland area. They are surrounded by tall green reeds and water. The sky is a mix of blue and orange, suggesting sunset or sunrise. The text is overlaid in the center of the image.

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APPENDIX: AUTOMATION



Unveiling the potential of automated processing and an all encompassing credit score

Cost and time savings via automation
(Automation leads to a 30% fall in the time taken to complete each step, on an average)

Total Time Taken Presently	70	Rationale: 7 hour workday	
Procedure	Cost Attributed	Time Spent(as a proportion of cost)	Savings in Time Spent
Loan Application	409	16.77211482	5.031634446
Online Form and Documentation	409	16.77211482	5.031634446
Automated Verification	267	10.94903339	3.284710018
Loan Eligibility	401	16.4440539	4.933216169
Auxiliary Services	221	9.06268307	2.718804921
Total	1707	70	21
Total Cost Savings	512.1	Rationale: No. of Hours saved* Hourly Costs	

Improvement in Financial Inclusion via Organic Growth of our Credit Reimagined Policy

Region	Population	Financially Excluded	Financially Inlcusion after Credit Reimagined
Sumatra	59977300	8936617.7	211524.7762
Java	151600000	22588400	534654.8789
Papua New Guinea	10329931	1539159.719	36431.05546
East Nusa Tangerra	5466285	814476.465	19278.20544
			801888.916
USA(Lending Circle	0.291666667		
India(SHG Model)	2.5		
Indonesia	2.97		
Weighted Average	2.366944444		

APPENDIX: LENDING CIRCLES

Exploring the mechanism



Lending Circles

Mobile Literate: Approximately 30%
Targeted through Online Applications

Lending Circles Application Form

Personal and Beneficiary Details



Name Adika
Age 28
Address 42, Java
Zip Code 6xxxx
Contact No. XXXXX XXXXX

Search Postal Code 

☐ Lending Circle 1
East Java, 6xxxx

☐ Lending Circle 2
Java, 1xxxx



Personal and Beneficiary Details

Lending Circles Application Form

Financial Details



Adika

Occupation Farmer
No. of Dependents 3
Annual Income 25,500,000
Past Loan Records 2

Search Postal Code 

☐ Lending Circle 1
East Java, 6xxxx

☐ Lending Circle 2
Java, 1xxxx



Financial Details

Less Aware about Mobile
Targeted through Agent Network



Agents or Field Officers visit the Tier 2/3 areas



Target groups individually after looking at synchronized data server



Awareness through pamphlets, on the spot application filling using Officer's device

APPENDIX: CREDIT SCORING

Unveiling the potential of automated processing and an all encompassing credit score



The Lending Circle Risk Score(P2P Credit Score)

Any default increases lending circle risk score which if pushed beyond a limit will **suspend loans**.
Acts as measure of Social Capital

Nominal Default	1 risk score addition
Wrong Information Provided	2 risk score addition

- Establishment of 0% interest initial lending circles aims at financial inclusion of low and middle income households
- Each circle would be required to furnish with details about the risk score of other members
- Large number of responses collected will remove any bias involved

Based on the final Credit Score, the following alternatives are recommended

LOW SCORE

ACCELERATED REPAYMENT

The loan repayment can be pushed a little earlier and at faster intervals, so that the overall risk is minimalised and spread out.

PROGRESSIVE LENDING

The loans can be lent out in a progressive manner with a **higher rate** starting from a smaller amount and then progressing towards larger sums to judge the credibility based on farmer's repayment.

HIGH SCORE

The person is granted the loan amount and the repayment would not be in a accelerated manner, but normalized manner



Adika, who struggled to get a loan earlier due to unavailability of information and low credit score is now granted a loan

APPENDIX: PRODUCT OFFERINGS

Exploring the attractiveness of Agriculture Value Chain Loans



Current situation of farmers...

Unfamiliar with loan products

Only **seven per cent** of farmers interviewed had taken out a loan from a formal institution.

Longer-Loan Terms

Both field agents and farmers prefer loan terms of **six months or more** to secure sufficient time for raising funds and repaying the loan.

Diverse needs for credit

Fifty-five per cent said they needed the loan for multiple reasons (crop diversification, non-farm expenses) while **45 per cent** said they would use the loan only for their cash crop.



...Existing Loan Products available to small farmers in Indonesia(focus on Papua)

PRODUCT	PARTNERS	VALUE CHAIN	TYPE OF LOAN	REGION	LAUNCHED	MONTHLY INTEREST
Agriculture Value Chain Loan (AVCL) ²⁵	MiBank, PNGAC, Cocoa Board, ADB, Pacific Private Sector Development Initiative (PPSDI)	Cocoa	Moveable asset: future contracted sales as collateral	East New Britain	2019	2%
Fisheries Loan ²⁶	MiBank, NDB, National Fisheries Authority of Papua New Guinea	Fish	Collateral equivalent to 30% of equity	Madang, Morobe	2019	1.5%
Speed Loan ²⁷	MiBank, Digicel	Various	Collateral in savings or deposit account	Nationwide	2019	2%-3%
Agricultural Loan	Women's Micro Bank, Kiva	Various	NA	Nationwide	NA	NA

Our Solution: Revamping the product on the basis of Bottlenecks

APPENDIX: CREDILINK

Choice of Account Aggregator and Key Savings through utilization of e-KYC



Choosing the Account Aggregator
[Finantier](#)

Choice of Account Aggregator

Account Aggregator	Ease of Use	Data cleanliness	API Offerings	Breadth of Connections
Finantier	✓	✓	✓	
FinFini	✓		✓	✓
Musoni		✓		✓
Oradien	✓		✓	

Data points used in the model from the study	Value (assumptions)
Average cost of the current process (including customer, merchants, borrowers, and lenders)	
a) E-money	IDR 24,000 (USD 1.62)
b) P2P	IDR 48,500 (USD 3.28)
Assumed cost of e-KYC through Dukcapil for one application	IDR 5,000-7000 (USD 0.34-0.47)
Current administrative costs	IDR 8,000
Assumed savings in administrative cost	<u>50%</u>

Data points used in the model	Value
Current number of customers:	
a) E-money	125 million
b) P2P	16.4 million
Growth rate for the e-money sector	<u>15%</u> for 5 years
Assumed mean revert rate	5%
Growth rate for the P2P sector	10% for 5 years
Assumed mean revert rate	2%
Population growth	1.01%
Interest-free rate used for present value calculations (across all years)	6.73%

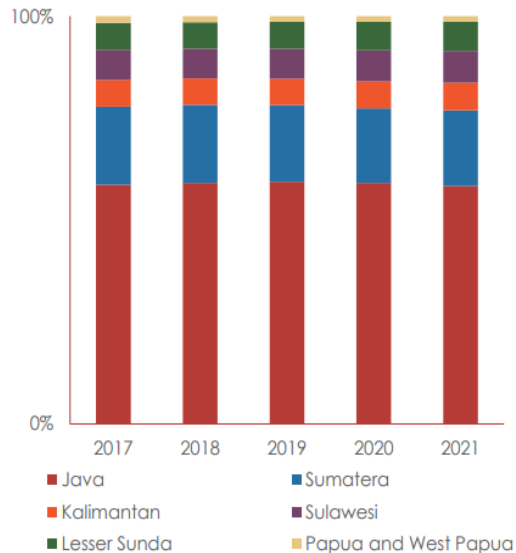
Credilink Impact: Assumptions

APPENDIX: Quantitatively Analysing Reasons for Expansion

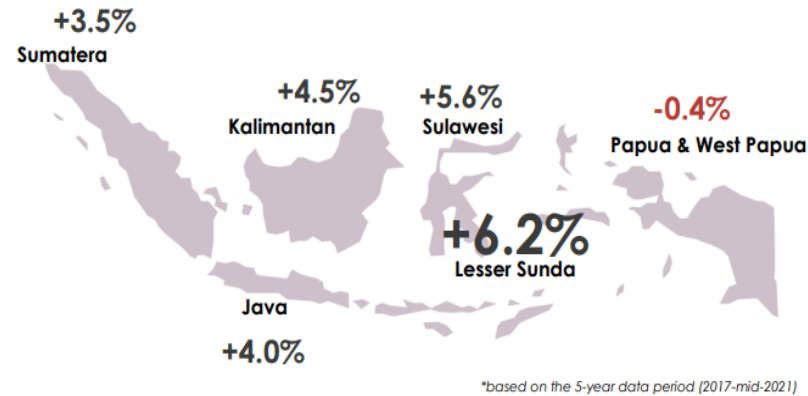
Exploring the untapped SME loan potential of East Nusa Tenggara



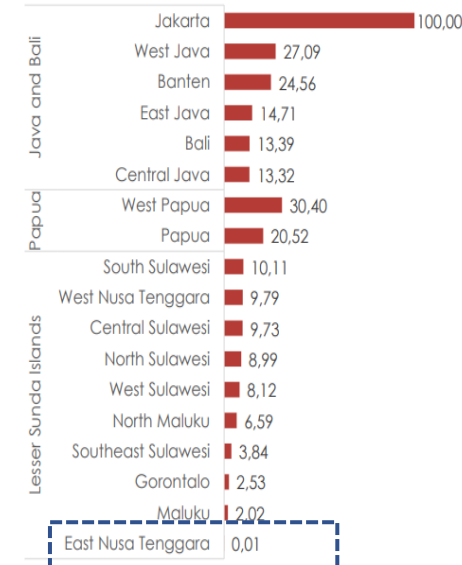
Around 59% of MSME loans are concentrated in Java



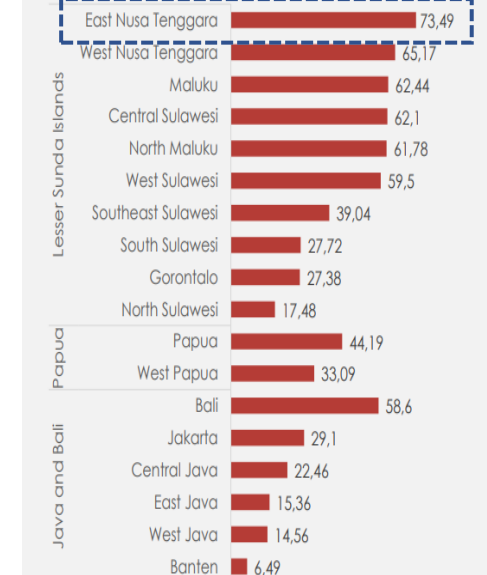
While the average annual growth* of MSME loans are the highest in the Lesser Sunda Islands



SME Productivity Index



SME Competitiveness Index



Unmet Demand and High Growth Potential

- Low disbursement of MSME loans
- Highest Annual Growth Rate of loans
- Lowest Productivity and Highest Competitiveness

preliminary clarifications for assessing targeted market growth rate



		ASSUMPTIONS								
Years	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
Base Case										
Growth Factor used to obtain market share					0%	0.00%	0.00%	0.20%	0.10%	0.10%
Net Increase in Loan Books(%)										
With Project Credilink					5.20%	5.73%	6.30%	6.94%	7.64%	8.41%
CAGR of Microfinance Industry(Indonesia)					10.10%					
(Assumption taken based on the industry cagr and pre-existing digital solution growth rates)										
Additional Investment Assumptions(y-o-y rate, % of Revenue)										
Maintenance						8.00%	8.00%	8.00%	8.00%	8.00% Average growth rate with respect to average loan size
Adoption and education - Marketing						7.00%	7.00%	7.00%	7.00%	7.00% Average growth rate with respect to average loan size
Platform						3.00%	3.00%	3.00%	3.00%	3.00% After inception, these reduce drastically
Set up costs							5.00%	5.00%	5.00%	5.00% Nominal for consecutive years after inception
Data Validation and analysis						8.00%	8.00%	8.00%	8.00%	8.00% More number of borrowers and more complex methods will increase costs in a sustained way
Salary Assumptions										
Employee Costs(% of total expenses)					79%	73%	71%	61%	53%	49%
(Employee costs would tend to decrease drastically owing to unemployment trends and recent layoffs)										
Total Expenses					2,192.00	2,025.54	2,066.41	2,057.60	2,077.91	2,083.13
Total Employee Costs					1,731.68	1,478.64	1,467.15	1,255.14	1,101.29	1,020.74

APPENDIX: CASH FLOW PROJECTIONS

Impact Assessment on the basis of Cash Flow Projections



Years	Cashflow Projections					
	2023A	2024E	2025E	2026E	2027E	2028E
Cashflow Projections(000,000's IDR)						
Loans and Advances						
Without Credilink	8,969.50	8,518.12	8,533.01	8,677.75	8,694.20	8,677.01
Project Credilink		9483.02	9055.05	9125.20	9340.82	9425.63
y-o-y growth(%)		5.73%	6.30%	6.94%	7.64%	8.41%
Net Interest Income Earned						
Without Project Credilink	1,995.86	1,895.42	1,898.74	1,930.94	1,934.60	1,930.78
Project Credilink		2,110.13	2,478.00	2,563.78	2,780.03	2,803.98
Increased Cashflows from lending		214.71	579.26	632.84	845.43	873.20
SALARY AND WAGES						
Project Credilink	1,731.68	1,478.64	1,467.15	1,255.14	1,101.29	1,020.74
Without Project Credilink	1,734.00	1,637.00	1,663.44	1,658.17	1,667.98	1,671.80
Cashflows from Costs Savings		158.36	196.29	403.04	566.69	651.07
Cashflow after adjusting for salaries and wages		373.07	775.55	1,035.87	1,412.11	1,524.27

The client breakdown given in the problem statement to find the total amount of loans disbursed in Jun 23 using flat rates and size of each loan

Calculated in assumptions with a weighted average sum of Indonesia's GDP and lending sector CAGR

APPENDIX: CASH FLOW PROJECTIONS

Impact Assessment on the basis of Cash Flow Projections



ADDITIONAL EXPENSES

Maintenance	143.00	265.07	286.27	309.17	333.91	360.62	
Transaction Costs		94.83	90.55	91.25	93.41	94.26	As per industry estimates, 1% of the total amount of disburseals is incurred in transaction costs
Merchant Discount Rate (MDR)		28.45	27.17	27.38	28.02	28.28	0.03% of total loans and advances
Technology Infrastructure Upgrade Fees		141.79	144.65	144.03	145.45	145.82	Global proxy used for technological expenditure(7%) in credit lending sector
Aggregator Charges		142.25	135.83	136.88	140.11	141.38	OneMoney India Account Aggregator and Data Analytics charges taken as proxy
Platform charges		152.02	156.58	161.28	166.12	171.10	Average Financial services app platform costs is \$5000/year in Indonesia
App development and functioning		152.02	156.58	161.28	166.12	171.10	
Set up costs		250.00	262.50	275.63	289.41	303.88	
Identification Layer		150.00	157.50	165.38	173.64	182.33	Based on team analysis and estimation of various components of each layer, a weightage of 60%(of total set-up costs) was given
Payment Layer		25.00	26.25	27.56	28.94	30.39	Based on team analysis and estimation of various components of each layer, a weightage of 10%(of total set-up costs) was given
Data Layer		45.00	47.25	49.61	52.09	54.70	Based on team analysis and estimation of various components of each layer, a weightage of 30%(of set-up costs) was given
Data Validation and analysis		184.92	176.57	177.94	182.15	183.80	
Data Validation(E-KYC setup)		9.48	9.06	9.13	9.34	9.43	assumed to be 0.001% due to Presence of E-KTM with 98% population leads to no onboarding or data collection c
Data Analytics cost		175.44	167.52	168.82	172.81	174.37	OneMoney India Account Aggregator and Data Analytics charges taken as proxy
Total		994.25	1,017.75	1,060.90	1,111.69	1,160.78	
NET CASH FLOW		-621.18	-242.20	-25.02	300.42	363.49	

APPENDIX: KEY CALCULATIONS

Important financial indicators to justify our strategy



NPV AND IRR CALCULATIONS					
	2024E	2025E	2026E	2027E	2028E
Year	1	2	3	4	5
Cashflows	-621.18	-242.20	-25.02	300.42	363.49
Present Value(P.V)	-574.05	-206.84	-19.75	219.11	244.99
Last Cashflow	363.49				
WACC	10.22%				
Value	3,556.63				
PV	1,983.70				
NPV	1,647.16				
FCF	-574.05	-206.84	-19.75	219.11	244.99
Terminal Value					1,983.70
Total	-574.05	-206.84	-19.75	219.11	2,228.69
IRR	36%				

COST OF EQUITY	
Risk-Free Rate	6.41%
Indonesia ERP	7.89%
Country Risk Premium	2.89%
Regression Beta	0.36
Cost of Equity(CAPM)	8.21%

(The financial structure only inculcates the cost of equity as **debt component is not present**)

(All measures taken from the website of **Aswath Damodaran**)

APPENDIX: CALCULATIONS FOR CREDILINK

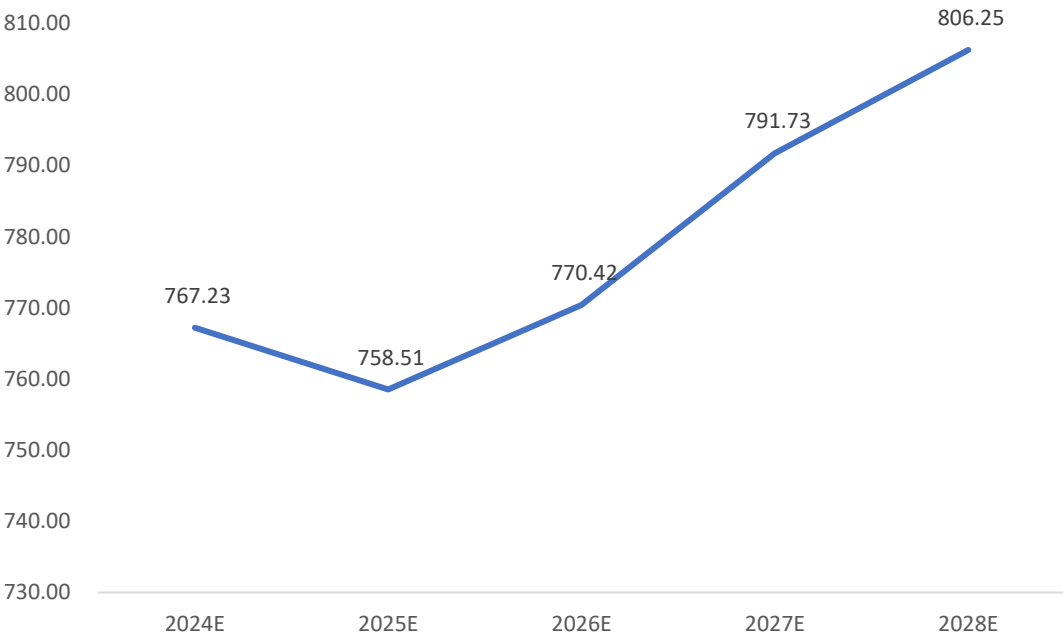
Justifying Credilink’s 3-pronged costing strategy



IDENTIFICATION LAYER					
	2024E	2025E	2026E	2027E	2028E
Setup Cost	150.00	157.50	165.38	173.64	182.33
Data Validation	9.48	9.06	9.13	9.34	9.43
Identifaction Layer Cost	159.48	166.56	174.50	182.98	191.75
PAYMENT LAYER					
	2024E	2025E	2026E	2027E	2028E
Transaction Costs	94.83	90.55	91.25	93.41	94.26
Merchant Discount Rate (MDR)	28.45	27.17	27.38	28.02	28.28
Technology Infrastructure Upgrade Fees	141.79	144.65	144.03	145.45	145.82
Setup Costs	25.00	26.25	27.56	28.94	30.39
Payment Layer Cost	290.07	288.61	290.22	295.83	298.74
DATA LAYER					
	2024E	2025E	2026E	2027E	2028E
Aggregator Charges	142.25	135.83	136.88	140.11	141.38
Data Analytics Charges	175.44	167.52	168.82	172.81	174.37
Setup Costs	45.00	47.25	49.61	52.09	54.70
Data Layer Cost	317.68	303.34	305.69	312.92	315.76
CREDILINK TOTAL PROJECT COST	767.23	758.51	770.42	791.73	806.25

Assumptions & Sources

1. Presence of E-KTM with 98% population leads to no onboarding or data collection cost
2. KYC Setup Cost -<https://seon.io/resources/cost-of-kyc/#h-3-implement-dynamic-friction-checks> = \$130 (= 2Mn IDR)
3. Authentication and E-KYC cost post setup - https://www.microsave.net/wp-content/uploads/2021/01/PB-24_-KYC-practices-in-Indon
4. Transaction Cost as per AyoConnect website, covers operational and other expenses
5. AyoConnect has waived off setup fees and disbursement integration fees for early birds
6. OneMoney India Account Aggregator and Data Analytics charges taken as proxy



APPENDIX: LOAN BREAKDOWN AND CLIENT OUTREACH



Assessing the effect of Credilink on credit lending in Indonesia

Loan Breakdown

LOAN BREAKDOWN(AS OF JUN-23)

Type of Loan	Number of Clients	Size(000's IDR)	Loans and Advances	Flat Rate(%)	Net Income Earned
Individual	1,051.00	3,500.00	36,78,500.00	22.50%	8,27,662.50
Small Enterprise	671.00	7,000.00	46,97,000.00	22.50%	10,56,825.00
Established Client	54.00	11,000.00	5,94,000.00	18.75%	1,11,375.00
Total			89,69,500.00		19,95,862.50

Client Outreach Projections

Years	2019A	2020A	2021A	2022A	CLIENT OUTREACH						Rationale
					2023A	2024E	2025E	2026E	2027E	2028E	
Loan applications											
Applied for loan											
With Credilink						340	351	362	374	386	
y-o-y growth(%)						3.24%	3.24%	3.24%	3.24%	3.24%	
Without Credilink	271	361	291	309	329	331	338	344	351	357	Forecasted by Regression Analysis
y-o-y growth(%)		33.21%	-19.39%	6.19%	6.47%	0.73%	1.93%	1.89%	1.86%	1.83%	
Year weight	1	2	3	4	5	6	7	8	9	10	
Accepted loans											
With Credilink						197	204	210	217	224	
y-o-y growth(%)						3.24%	3.24%	3.24%	3.24%	3.24%	
Without Credilink	195	191	203	178	191	185.3	183.2	181.1	179	176.9	
y-o-y growth(%)		-2.05%	6.28%	-12.32%	7.30%	-2.98%	-1.13%	-1.15%	-1.16%	-1.17%	

APPENDIX: MARKET PENETRATION

Estimating Market Size with Phoenix



CLIENT ESTIMATION IN THE POTENTIAL STATES					
In Mn					Rationale
Population Divide	Java	Sumatra	Papua	East Nusa Tenggara	
Total Population	151.6	59.98	4.42	5.47	2022 Official Estimate - Wiki
Males(%)	49.60%	49.95%	49.44%	50.10%	Determined from sex ratio calculations
Females(%)	50.40%	50.05%	50.56%	49.90%	Determined from sex ratio calculations
Males	75.19	29.96	2.18	2.74	
Females	76.41	30.02	2.23	2.73	
Urban-Rural Divide					
Urban Population(% of total population)	56.60%	56.20%	62.20%	62.20%	https://www.adb.org/sites/default/files/publication/26436/informal-sector-indonesia.pdf
Rural Population(% of total population)	43.40%	43.80%	37.80%	37.80%	https://www.adb.org/sites/default/files/publication/26436/informal-sector-indonesia.pdf
Rural Population	65.8	26.3	0.8	2.1	
Occupation Divide					
% people engaged in agriculture(% of Rural Population)	63.50%	57.70%	67.20%	45.20%	World bank Estimates - Assumption of % as Indonesia's employment
% people not engaged in agriculture(% of Rural Population)	36.50%	42.30%	32.80%	54.80%	World bank Estimates - Assumption of % as Indonesia's employment
People Engaged in Agriculture	41.8	15.2	0.5	0.9	
Informal sector(%)	70.00%	57.00%	70.90%	61.20%	World bank Estimates - Assumption of % as Indonesia's employment
Informal sector	29.2	8.8	0.4	0.6	
Formal sector(%)	30.00%	42.20%	29.10%	38.80%	World bank Estimates - Assumption of % as Indonesia's employment
Formal sector	12.5	6.4	0.2	0.4	
People not Engaged in Agriculture	24.0	11.1	0.3	1.1	
NECESSATY BASIS					
Farmers that require Credit	16.7	6.1	0.2	0.4	Based on Survey analysis, 40% require credit
Non-Farmers that require Credit	4.80	2.22	0.05	0.23	Based on Survey analysis, 20% require credit
Total	21.5	8.3	0.3	0.6	
Intended market share(%)	10.00%	10.00%	10.00%	10.00%	Based on Team estimates accounting for similar proportion in lending sector
Market size to capture	2.15	0.83	0.03	0.06	
Conversion Ratio(%)	0.20%	0.20%	0.20%	0.20%	Based on Team estimates accounting for similar proportion in lending sector
Potential Clientele(Number of Clients)	4303	1657	53	120	

APPENDIX: FAILED INITIATIVE - KUR

Analysing Micro KUR requirements



Micro KUR requirements (according to BRI):
- Individuals who carry out productive and viable businesses
- Has been actively doing business for at least 6 months
- Not receiving credit from banks except for consumer loans such as KPR (Home Ownership Loans), KKB (Motor Vehicle Loan), and credit cards
- Administrative requirements: Identity in the form of KTP (Indonesian identity card), Family Card (KK), and business permit
Micro KUR's terms:
- Maximum loan of IDR 50 million/debtor
- Loan types: Working Capital Loan (KMK) with a maximum term of three years/ Investment Loan (KI) with a maximum term of five years
- Effective interest rate of 6% per year
- No provision fee or administration fee
- No collateral required
KUR requirements (according to BRI):
- Have a productive and productive business
- Not receiving credit from banks except for consumer loans such as KPR, KKB, and Credit Cards
- Has been actively doing business for at least 6 months
- Have a Micro and Small Business Permit (IUMK) or other business permit that can be equalized
KUR terms:
- Loan of IDR 50 – 500 million
- Loan types: Working Capital Loan (KMK) with a maximum term of three years; Investment Credit (KI) with a maximum term of five years
- Effective interest rate of 6% per year
- No provision fee or administration fee
- Collateral per the bank's regulation

APPENDIX: REFERENCES



The sources we referred to

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Thank you